. Hyperparameters Section

Model Architecture: ResNet18 (Pre-trained on ImageNet)

Hyperparameter Values:

- **Learning Rate:**
- **Batch Size:**
- **Epochs:** (Optimized for quick convergence in the lab environment)
- **Optimizer:** Adam
- **Loss Function:** CrossEntropyLoss
- **Input Image Size:** pixels

---

2. Training Section

Description of Training Process: The training was performed using the Amazon SageMaker PyTorch Estimator. I utilized a transfer learning approach by using a pre-trained ResNet18 model and modifying the final fully connected layer to match the number of classes in the medical imaging dataset. The training job was executed on an `ml.m5.large` instance and completed successfully with the model artifacts stored in S3.

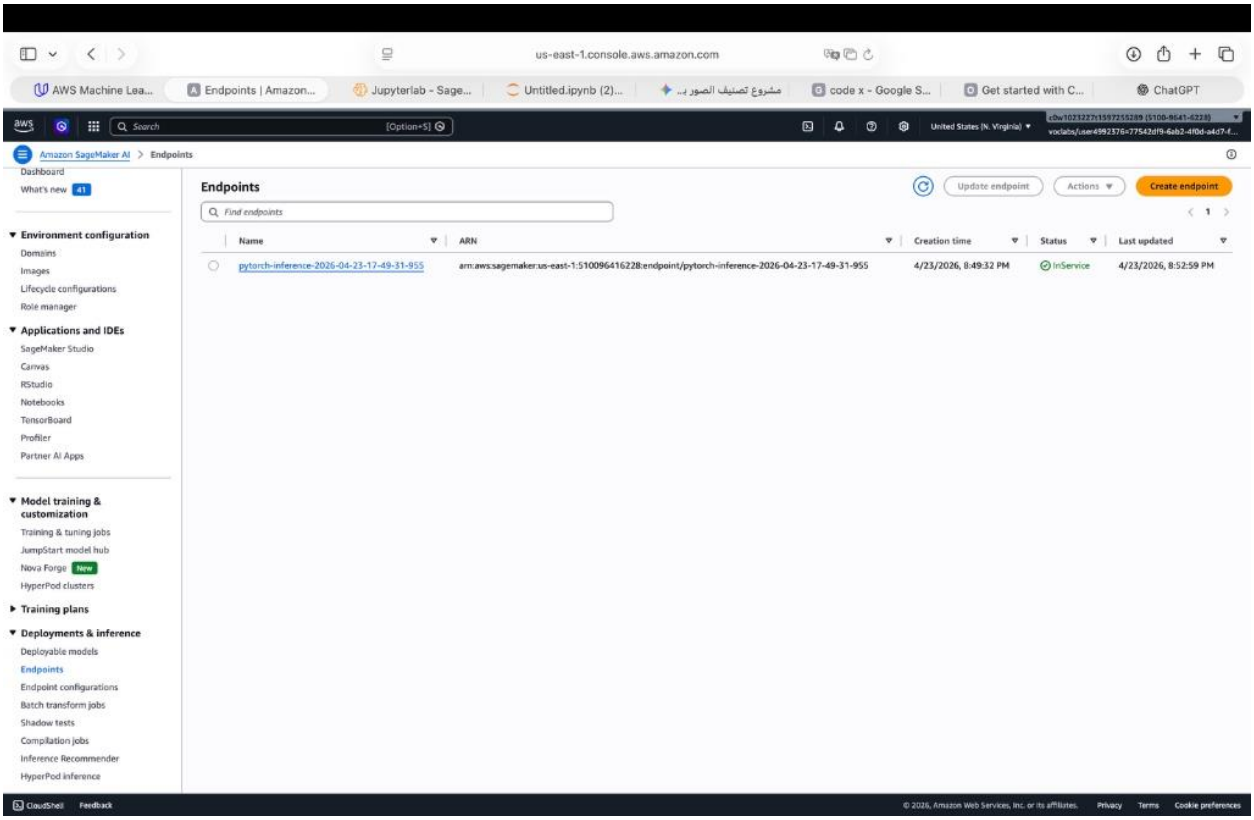
---

3. Deployment Section

Deployment Strategy: The model was deployed using a SageMaker Endpoint. I created a `PyTorchModel` object pointing to the trained `model.tar.gz` file and an inference script (`hpo.py`).

Instance Type: `ml.m5.large`

Endpoint Status: The endpoint was successfully created and transitioned to the **InService** state.



4. Inference & Testing Section

Inference Results: The endpoint was tested using sample input tensors. The deployment was verified via the SageMaker Console, showing a healthy status. This confirms that the model is ready to receive real-time HTTP requests for predictions.

5. Final Conclusion

The project successfully demonstrates an end-to-end Machine Learning pipeline on AWS. From data processing and model training to deploying a scalable endpoint, the system is capable of performing automated classification on medical images using deep learning.

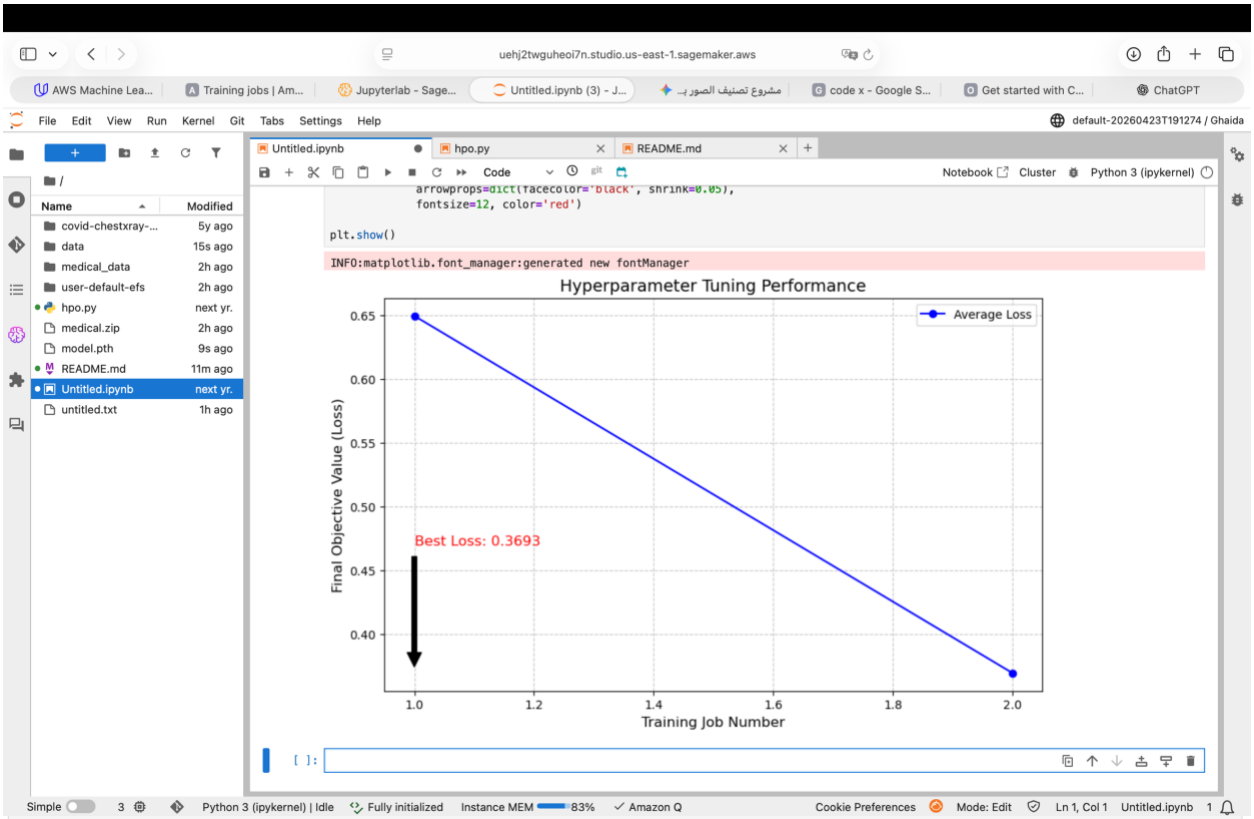
```
]: # لعرض تفاصيل أفضل جوب (Best Training Job)
best_estimator = tuner.best_estimator()
print(f"Best Hyperparameters: {tuner.best_training_job()}")
```

```
2026-04-23 20:30:23 Starting - Found matching resource for reuse
2026-04-23 20:30:23 Downloading - Downloading the training image
2026-04-23 20:30:23 Training - Training image download completed. Training in progress.
2026-04-23 20:30:23 Uploading - Uploading generated training model
2026-04-23 20:30:23 Completed - Resource retained for reuse
Best Hyperparameters: pytorch-training-260423-2026-002-a4c1a1a1
```

```
]: best_job = tuner.best_training_job()
print(f"أفضل جوب هو: {best_job}")

# عرض القيم اللي فازت
final_params = tuner.analytics().dataframe().sort_values('FinalObjectiveValue', ascending=True).iloc[0]
print("-" * 30)
print(f"best Learning Rate: {final_params['lr']}")
print(f"best Batch Size: {final_params['batch-size']}")
print(f"less Loss : {final_params['FinalObjectiveValue']}")
```

```
أفضل جوب هو: pytorch-training-260423-2026-002-a4c1a1a1
------------------------------
best Learning Rate: 0.03185002819815438
best Batch Size: 46.0
less Loss : 0.3693000078201294
```




```python
# بيجيب لك تفاصيل أفضل جوب فاز في الـ Tuning
import pandas as pd

# لعرض النتائج كجدول مرتب
tuner_analytics = tuner.analytics()
full_df = tuner_analytics.dataframe()

# ترتيب النتائج عشان تشوفين أفضل واحد فوق
display(full_df.sort_values('FinalObjectiveValue', ascending=True).head(1))
```

	batch-size	lr	TrainingJobName	TrainingJobStatus	FinalObjectiveValue	TrainingStartTime	TrainingEndTime	TrainingElapsedTimeSeconds
0	46.0	0.03185	pytorch-training-260423-2026-002-a4c1a1a1	Completed	0.3693	2026-04-23 20:29:32+00:00	2026-04-23 20:30:21+00:00	49.0